



Additional Practice Questions

Capacitance

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4											
C	D	B	A											

- 1 C Total charge on C_1 is $Q = C_1 V = 100C_1$

When C_1 and C_2 are joined in parallel, the charges on the 2 capacitors are

$$Q_1 = 60C_1 \text{ and } Q_2 = 60C_2$$

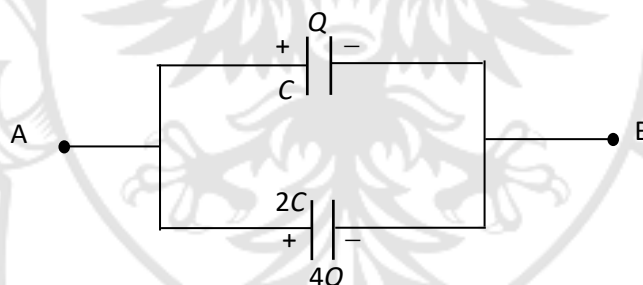
Since $Q_1 + Q_2 = 60C_1 + 60C_2 = 100C_1$

$$40C_1 = 60C_2$$

$$\therefore \frac{C_1}{C_2} = \frac{60}{40} = \frac{3}{2}$$

- 2 D When the two charged capacitors are joined with their positive plates connected together and their negative plates connected together, charge will redistribute until the p.d. across both capacitors is the same. This means there will be a current I flowing in the circuit. Thus, there will be energy loss due to Joule heating. The total charge in the system is conserved.

Proof: (Not required in the answer).



For capacitor C_1 ,

$$Q_1 = CV$$

For capacitor C_2 ,

$$Q_2 = 2C \times 2V = 4CV$$

$\therefore Q_2 = 4Q_1 \Rightarrow Q_{Total} = 5Q_1 = 5CV$ and this is conserved.

When the two capacitors are connected in parallel, charge will flow from C_2 to C_1 until the p.d. across both capacitors is the same

Initial total energy stored in the capacitors is

$$= \frac{1}{2}CV^2 + \frac{1}{2}(2C)(2V)^2 = \frac{9}{2}CV^2$$

Final total energy stored is

$$\frac{1}{2} \frac{Q_{total}^2}{3C} = \frac{1}{2} \frac{(5CV)^2}{3C} = \frac{25}{6}CV^2$$

Since $\frac{25}{6}CV^2 < \frac{9}{2}CV^2$ this means energy is lost.

- 3 B** Energy dissipated in the tube is = energy lost by capacitor

$$= \frac{1}{2}CV_2^2 - \frac{1}{2}CV_1^2 = \frac{1}{2}C(V_2^2 - V_1^2) = \frac{1}{2} \times 160 \times 10^{-6} (200^2 - 100^2) \\ = 2.4 \text{ J}$$

- 4 A** Consider the capacitors in parallel, p.d. V is the same across the top and bottom branch

Charge stored in the top branch's capacitor

$$Q_{top} = CV$$

Charge stored in the bottom branch's capacitor

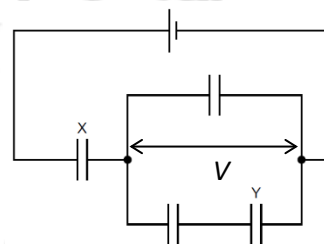
$$Q_{bottom} = \left(\frac{1}{C} + \frac{1}{C} \right)^{-1} V = \frac{C}{2} V$$

Capacitors in series store the same magnitude of charge

$$Q_Y = Q_{top} + Q_{bottom}$$

$$6 = 2Q_{bottom} + Q_{bottom}$$

$$Q_{bottom} = 2 \mu\text{F}$$

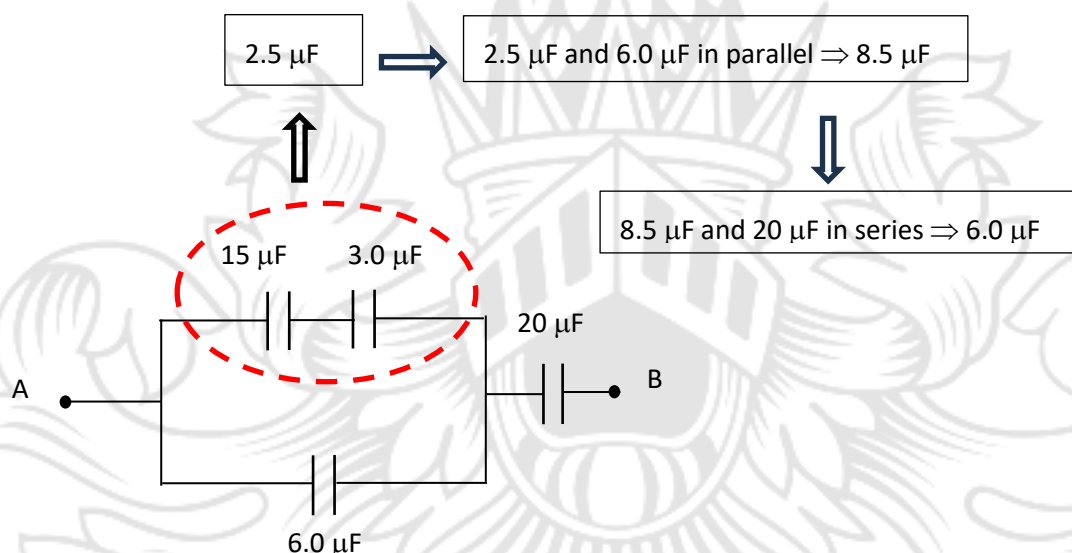
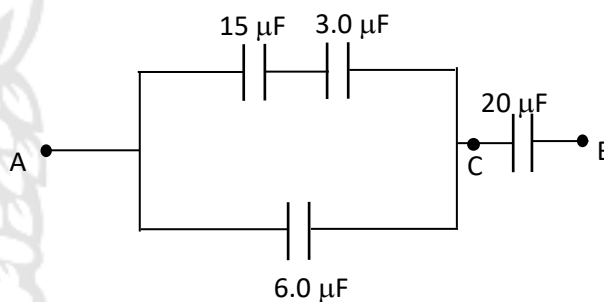


II. Structured Questions

1 a.

$$C_{AC} = \left(\frac{1}{15} + \frac{1}{3} \right)^{-1} + 6 = 8.5 \mu\text{F}$$

$$C_{AB} = \left(\frac{1}{8.5} + \frac{1}{20} \right)^{-1} = 6.0 \mu\text{F}$$



- b. Since $V_{AB} = 15.0 \text{ V}$, $Q = CV = 6.0 \times 15 = 90 \mu\text{C} \Rightarrow$ charge on $20 \mu\text{F}$ is $90 \mu\text{C}$. (capacitors in series store the same magnitude of charge)

Charge on the equivalent capacitor C_{AC} is also $90 \mu\text{C}$.

$$V_{AC} = \frac{90}{8.5} = 10.6 \text{ V}$$

$$Q_{6.0 \mu\text{F}} = 6.0 \times 10.6 = 63.6 \mu\text{C}$$

Q across the equivalent capacitors of $15 \mu\text{F}$ and $3.0 \mu\text{F}$ in series,

$$Q_{2.5 \mu\text{F}} = 2.5 \times 10.6 = 26.5 \mu\text{C}$$

Charge on both $15 \mu\text{F}$ and $3.0 \mu\text{F}$ are also $26.5 \mu\text{C}$.

- 2** Initial charge on C_1 is $6.0 \times 20 = 120 \mu\text{C}$

When S_2 is closed, the total charge of $120 \mu\text{C}$ is redistributed between C_1 and C_2 until the p.d. across both capacitors are the same.

$$Q = Q_1 + Q_2 = C_1 V + C_2 V = 6.0V + 3.0V = 120 \mu\text{C}$$

$$\therefore V = \frac{120}{9.0} = 13.3$$

$$Q_1 = C_1 V = 6.0 \times 13.3 = 80 \mu\text{C}$$

$$Q_2 = C_2 V = 3.0 \times 13.3 = 40 \mu\text{C} \quad \text{OR}$$

$$\text{Since charge is conserved, } Q_2 = 120 - 80 = 40 \mu\text{C}$$

- 3 a.** p.d. across $15 \text{ k}\Omega$ resistor

$$V_{15\text{k}\Omega} = \frac{9.0 \times 15}{12 + 15} = 5.0 \text{ V}$$

Initial charge on capacitor

$$Q = CV = 10 \times 5.0 = 50 \mu\text{C}$$

- b i.** When switch is opened, the fully charged capacitor will discharge through a total resistance of $15 + 3.0 \text{ k}\Omega = 18 \text{ k}\Omega$.

Therefore, the time constant of the circuit

$$\tau = RC = 18 \times 10^3 \times 10 \times 10^{-6} = 180 \text{ ms}$$

$$I = I_0 e^{-t/RC}$$

$$I_0 = \frac{5.0}{18 \times 10^3} = 2.78 \times 10^{-4} \text{ A}$$

$$I = 2.8 \times 10^{-4} e^{-t/180 \times 10^{-3}}$$

- ii.** During the discharge of the capacitor,

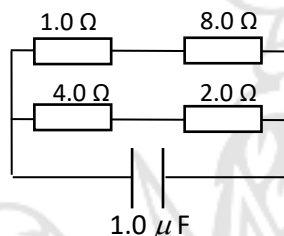
$$V_{\text{resistor}} = V_{\text{capacitor}}$$

$$\therefore V = IR = \frac{Q}{C} \Rightarrow \frac{IR}{I_0 R} = \frac{Q/C}{Q_0/C} = \frac{Q}{Q_0} = 0.20$$

$$I = I_0 e^{-t/RC} \Rightarrow \frac{I}{I_0} = e^{-t/RC} = 0.20$$

$$\ln\left(e^{-t/180 \times 10^{-3}}\right) = \ln 0.20 \Rightarrow t = 290 \times 10^{-3} \text{ s}$$

- 4 a. By potential divider principle:
 p.d. across the $1.0\ \Omega$ resistor is 2.0 V , and
 p.d. across the $8.0\ \Omega$ resistor is 8.0 V .
 Hence p.d. across the capacitor is $8.0 - 2.0 = 6.0\text{ V}$
- b. When switch is opened, the circuit looks like this:



The capacitor is discharging through an equivalent resistance

$$R_{eq} = \frac{(9.0)(6.0)}{(9+6)} = 3.6\ \Omega$$

$$V = V_0 e^{-t/\tau} \Rightarrow \frac{V}{V_0} = e^{-t/\tau} = \frac{1}{10}$$

$$\ln\left(\frac{V}{V_0}\right) = -t/\tau \Rightarrow t = \tau \ln\left(\frac{V_0}{V}\right)$$

$$\therefore t = 3.6 \times 1.0 \times 10^{-6} \ln(10) = 8.289 \approx 8.3\ \mu\text{s}$$

- 5 a. Capacitance is defined as the ratio of the magnitude of the charge stored on either conductor to the magnitude of the potential difference between the conductors.

b i. $Q = Q_0(1 - e^{-t/RC})$

$$0.95 = (1 - e^{-t/RC})$$

$$t = -RC \ln(1 - 0.95)$$

$$= -50 \times 120 \times 10^{-6} \ln(1 - 0.95) = 0.0180\text{ s}$$

- ii. Charge on the capacitor

$$Q = 0.95CV = 0.95(120 \times 10^{-6})(240) = 0.0274\text{ C}$$

Max energy delivered to the flash circuit

$$U = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} \left(\frac{0.0274^2}{120 \times 10^{-6}} \right) = 3.13\text{ J}$$